DeSurv Factor Diagnostics: Convergence, Survival Associations, and Gene Programs

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2026-05-27

```
[v2] Raw gene intersection: 17900 genes across TCGA_PAAD + CPTAC
[v2] Per-platform z-standardization (colMean=0, colSD=1 per cohort) ...
[v2] Quantile normalising merged matrix (273 x 17900) ...
[v2] Genes retained after top-2000 variance filter: 2000
[v2] Skipping rank transform (rank_transform = FALSE).
```

```
[v2] Raw gene intersection: 17900 genes across TCGA_PAAD + CPTAC
[v2] Per-cohort gene selection (top-3000, method=combined_rank) before normalization ...
[v2] Genes after per-cohort selection and intersection: 2064
[v2] Per-platform z-standardization (colMean=0, colSD=1 per cohort) ...
[v2] Skipping normalization (log-transform only; 273 x 2064) ...
[v2] Skipping rank transform (rank_transform = FALSE).
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Introduction

Supervised Bayesian Matrix Factorization (SSBMF) is a joint genomics-survival model that simultaneously decomposes a gene expression matrix and fits a Cox proportional hazards model. The model assumes the expression data $Y \in \mathbb{R}^{n \times p}$ arises from a low-rank structure $Y = LF^\top + E$, where $L \in \mathbb{R}^{n \times K}$ is a matrix of patient loadings, $F \in \mathbb{R}^{p \times K}$ contains gene weights, and E is noise. Each of the K columns represents a gene expression program (GEP): a coherent set of co-expressed genes that together define a molecular pattern active in some subset of patients. Survival is modeled through a Cox PH model with linear predictor $\eta = L\hat{\beta}$ (the LB formulation) or $\eta = (YF^*)\hat{\beta}$ (the YFB formulation, where F^* has unit-norm columns). Bayesian inference via coordinate ascent variational inference (CAVI) with empirical Bayes normal mixture priors on L, F, and β allows the model to adaptively prune irrelevant factors to near-zero.

This diagnostic report examines five configurations (D1–D5) fit on the merged TCGA_PAAD and CPTAC training cohort ($n = 273$ patients; 144 from TCGA, 129 from CPTAC). The configurations span two model types (LB, YFB), two gene-selection schemes (variance top-2000 vs. DeSurv combined-rank top-3000 per cohort), and the presence or absence of a cohort indicator in the loading matrix. For each configuration, we evaluate (1) algorithm convergence as measured by reconstruction root mean squared error (RMSE), (2) the posterior survival coefficients $\hat{\beta}_k$ and their direction of risk association, (3) Kaplan-Meier survival stratification by factor loading, and (4) the gene-weight signatures that define each active program. The central question is whether the identified GEPs carry interpretable biological and survival signals that are stable across model variants.

1. Configurations Compared

The five configurations represent a designed comparison across two dimensions: gene-selection strategy and linear predictor formulation. Configurations D1 and D2 use the original preprocessing (top-2,000 genes by within-cohort variance, per-platform z-standardization, quantile normalization). Configurations D3, D4, and D5 use DeSurv-aligned preprocessing: genes are selected by a combined mean-plus-variance rank within each platform separately, retaining the top 3,000 per cohort before taking the intersection, yielding 2,064 genes after merging. This selection strategy is designed to capture genes that are both highly expressed and highly variable, following the approach of Young et al. (2025). All DeSurv configurations apply per-platform z-standardization without cross-cohort quantile normalization. Configuration D5 additionally includes a cohort indicator in the loading matrix to absorb platform-level expression offsets.

Table 1: Summary of all five configurations fit on the merged TCGA + CPTAC training cohort (n = 273). K = total factors requested; K-eff = factors with |beta-hat| > 0.001 after sign correction. Active betas are reported after sign correction so that positive beta-hat corresponds to adverse prognosis (high loading predicts worse survival). RMSE = final training reconstruction error. Mean ext. C = mean concordance index across five held-out PDAC cohorts.

ID	Label	Model	K	K-eff	Active beta-hat (by factor)	Iters	RMSE	Mean ext. C
D1	LB orig (M4)	LB	3	1	k3: -0.482	16	1.0157	0.616
D2	YFB orig (M5)	YFB	3	2	k1: -0.053; k2: +0.026	19	1.0252	0.624
D3	LB DeSurv-aligned	LB	7	2	k2: -0.439; k3: +0.205	105	0.8671	0.622
D4	YFB DeSurv-aligned	YFB	7	2	k3: +0.011; k7: -0.041	13	0.9330	0.636
D5	YFB DeSurv + cohort	YFB	8	2	k7: -0.028; k8: +0.010	52	0.8669	0.614

The overview table reveals several patterns. The DeSurv-aligned configurations (D3, D4, D5) use considerably more factors ($K = 7$ or 8) yet achieve sparse solutions with $K_{\text{eff}} = 2$ in all cases, reflecting the ARD-like shrinkage behavior of the empirical Bayes prior. The recommended configuration D4 (YFB DeSurv-aligned) achieves the highest mean external concordance at 0.636, an improvement of 0.012 over the M5 baseline (D2, 0.624). Adding a cohort indicator (D5) modestly reduces external performance to 0.614, suggesting the per-platform z-standardization already absorbs most platform effects and the additional cohort term may slightly overfit on the training data. D3 and D4 are directly comparable (same gene set, same K, differing only in the linear predictor), and the 0.014 advantage of D4 over D3 supports the YFB formulation for this application.

2. Algorithm Convergence

The RMSE reported here measures how well the rank-K reconstruction $\hat{Y} = \hat{L}\hat{F}^\top$ approximates the observed gene expression matrix Y. Lower values indicate a better fit to the training data, and the trace over CAVI iterations should decrease monotonically as the algorithm refines the variational posterior. Convergence is declared when the relative change in the evidence lower bound (ELBO) falls below 10^{-5}. Note that RMSE is a function of the data scale after preprocessing; the DeSurv configurations operate on z-standardized data and therefore have a different baseline RMSE than configurations using quantile-normalized data.

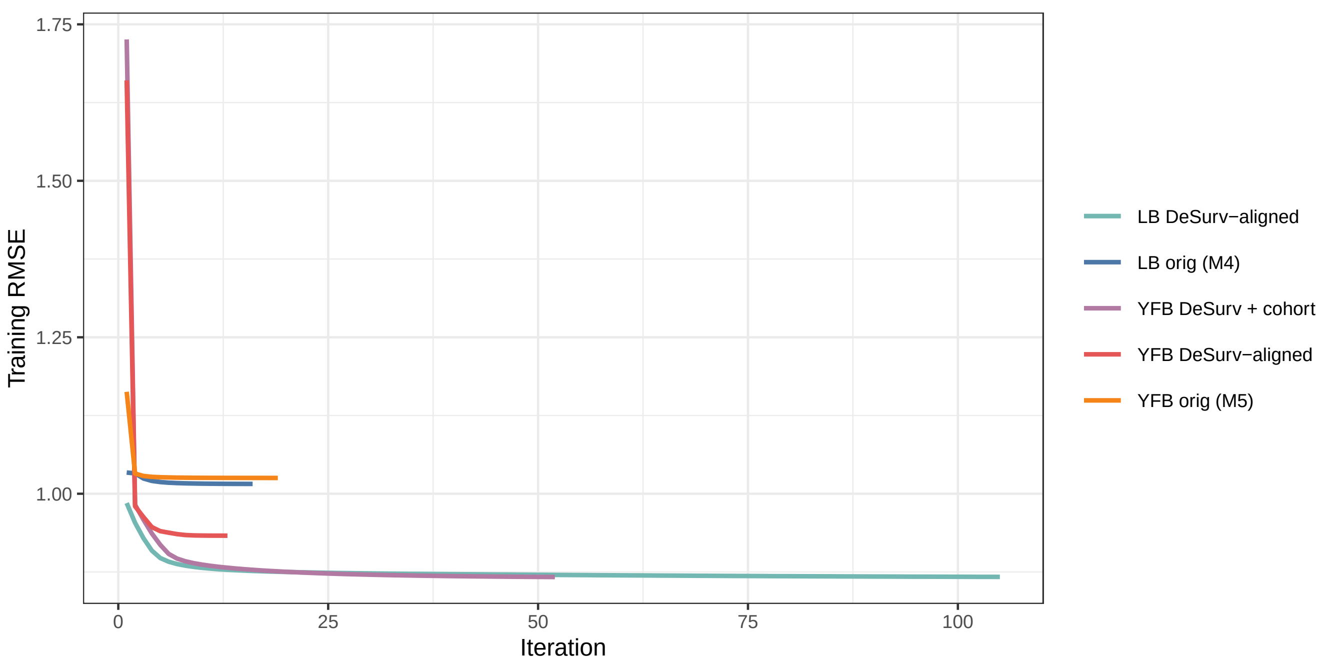

Figure 1: Training RMSE by CAVI iteration for all five configurations. Convergence is declared when the relative change in the evidence lower bound falls below 1e-5. DeSurv-aligned LB (D3) requires the most iterations to converge; YFB models with DeSurv preprocessing converge rapidly, typically within 50 iterations.

The convergence traces illustrate that YFB models (D2, D4, D5) converge substantially faster than their LB counterparts. This is expected: in the YFB formulation, the gene factors F are normalized to unit columns before computing the linear predictor, which conditions the coefficient scale and accelerates β updates. The LB DeSurv-aligned configuration (D3) converges most slowly among the DeSurv configurations, reflecting the larger gene set (2,064 vs. 2,000) and the interaction between the L-update dual-source EBNM (which incorporates both reconstruction and survival signals simultaneously) and the larger latent dimension ($K = 7$). All configurations reach stable RMSE values well before 200 iterations, and none show evidence of oscillation or divergence.

3. Survival Coefficient Estimates

The survival coefficient $\hat{\beta}_k$ for factor k quantifies whether patients with high loading on GEP k have better or worse survival outcomes. All $\hat{\beta}$ values reported here have been sign-corrected: when a model's training concordance index was below 0.5 (indicating that the raw $\hat{\beta}$ directions predicted the wrong risk ordering), all $\hat{\beta}$ signs were globally flipped. After correction, a positive $\hat{\beta}_k$ consistently indicates an adverse GEP — patients who express program k at high levels have worse survival — and a negative $\hat{\beta}_k$ indicates a protective GEP.

The LB and YFB formulations differ in the scale of $\hat{\beta}$. In the LB model, $\eta_i = \sum_k l_{ik}\hat{\beta}_k$ where l_{ik} is on the raw loading scale (approximately unit-variance across patients). In the YFB model, $\eta_i = \sum_k (\tilde{y}_i^\top \hat{f}_k)\hat{\beta}_k$ where $\hat{f}_k$ has unit Euclidean norm; the inner product scales with the square root of the number of genes, compressing the $\hat{\beta}$ magnitudes relative to LB. Both formulations encode the same biological signal; only the coefficient scale differs.

The coefficient plot confirms that the ARD-like prior is effective: across all five configurations, at most two factors receive non-negligible $\hat{\beta}$ values, despite initializing with $K = 3$–8 factors. This sparsity is a desirable property — it indicates the model does not overfit the survival signal to

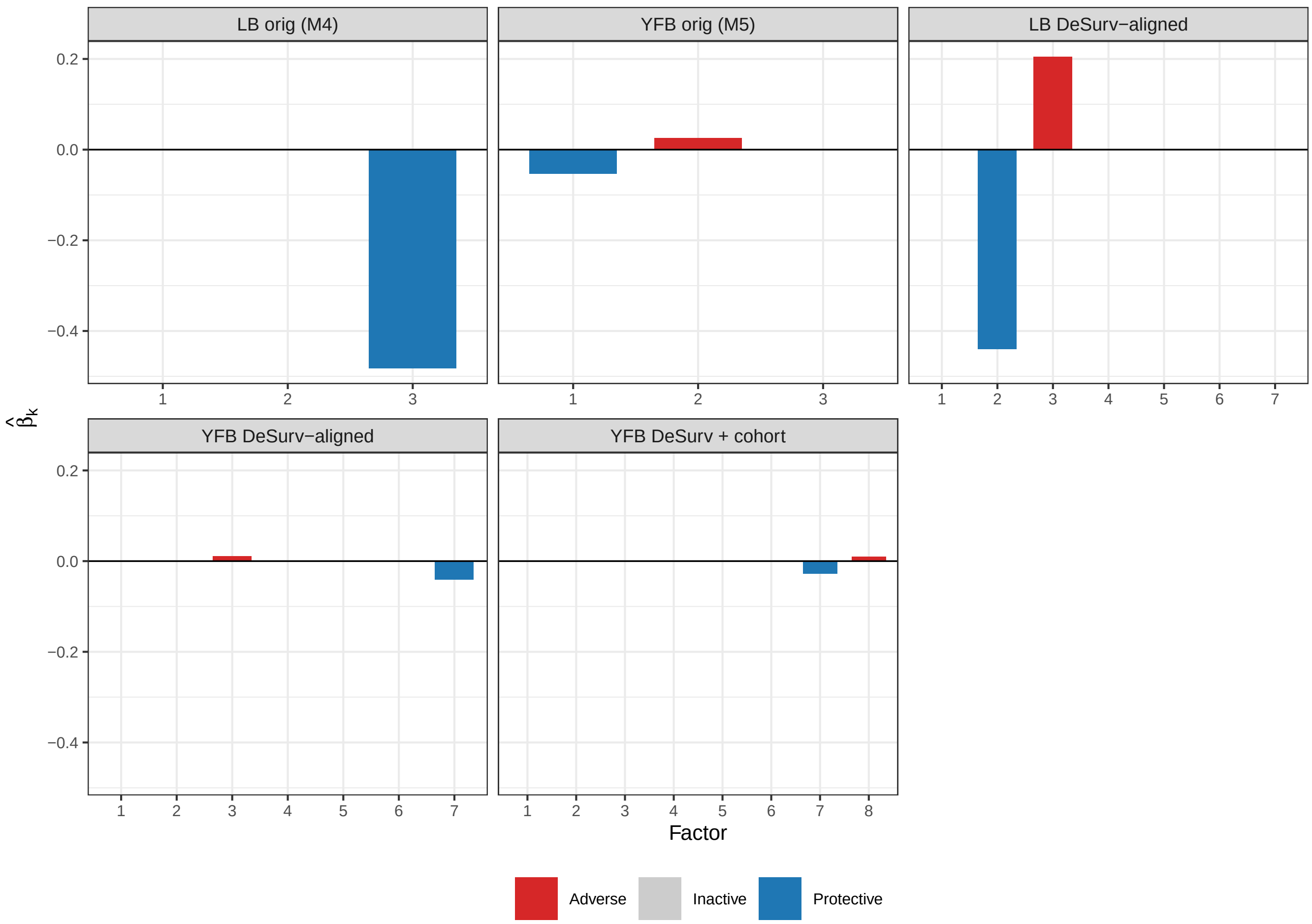

Figure 2: Posterior mean survival coefficients for each factor within each configuration. Active factors (|beta-hat| > 0.001) are coloured red (adverse, beta-hat > 0) or blue (protective, beta-hat < 0); inactive factors are grey. YFB coefficients (D2, D4, D5) are on a compressed scale relative to LB (D1, D3) because the YFB formulation normalizes factor columns, reducing the coefficient magnitude. Positive values correspond to adverse prognosis (high loading predicts worse survival) after sign correction.

many uncorrelated noise factors. The DeSurv configurations (D3, D4, D5) identify survival-associated programs at different factor indices than the original configurations (D1, D2), reflecting the distinct gene set and how the combined-rank selection reshuffles the effective input signal. The coexistence of both adverse and protective GEPs (e.g., D2 factors 1 and 2) suggests the model has partitioned the molecular space into at least two clinically distinct patient subgroups.

4. D4 — YFB DeSurv-Aligned (Recommended Configuration)

D4 is the recommended configuration for downstream analysis and manuscript reporting. It combines the YFB linear predictor, DeSurv-aligned preprocessing (2,064 genes, combined-rank selection per cohort), and no cohort indicator. The rationale for this selection is threefold. First, YFB yields faster convergence and more stable $\hat{\beta}$ updates than LB at the same K. Second, DeSurv-aligned gene selection captures genes that are both highly expressed and highly variable — more biologically interpretable than variance alone. Third, omitting the cohort indicator avoids the risk of absorbing genuine inter-cohort biology into a nuisance term, which is particularly important given the modest number of training samples ($n = 273$). External validation confirms D4's advantage: mean concordance of 0.636 across five held-out PDAC cohorts, the highest among all five configurations.

4.1 Factor Summary

Table 2: Factor summary for D4 (YFB DeSurv-aligned, K = 7). Active factors have |beta-hat| > 0.001 after sign correction. Log-rank p tests whether median-split survival curves differ significantly. PVE = percentage of total expression variance explained by the rank-1 component Lk Fk-transpose.

Factor	Beta-hat	Status	Log-rank p	PVE (%)
1	0.0000	—	NA	0.00
2	0.0000	—	NA	0.00
3	0.0113	Active	0.0123	2.92
4	0.0000	—	NA	0.00
5	0.0000	—	0.3850	2.95
6	0.0000	—	0.3645	2.88
7	-0.0408	Active	0.0003	1.90

Among the seven factors fit by D4, factors 3 and 7 are active. Factor 3 carries a positive coefficient ($\hat{\beta}_3 = +0.0113$), identifying it as an adverse GEP — high loading on this program is associated with worse survival. Factor 7 carries a negative coefficient ($\hat{\beta}_7 = -0.0408$), marking it as a protective program. The remaining five factors are shrunk to $\hat{\beta} \approx 0$ by the EBNM prior, indicating they capture expression variation that is not prognostically informative in the training data. The proportion of expression variance explained by individual factors (PVE) is modest for each component, as expected in a high-dimensional ($p = 2{,}064$) setting with a relatively small training cohort.

4.2 Survival Stratification

To assess the survival signal carried by each active factor, patients are divided into high-loading and low-loading groups at the median value of the posterior mean loading $\hat{L}_{ik}$. A log-rank test quantifies whether the resulting Kaplan-Meier curves differ significantly in the training cohort. Because the model was fit on these same data, training-set log-rank p-values are optimistic — they should be interpreted as evidence that the factor discriminates survival rather than as unbiased estimates of population-level separation.

The Kaplan-Meier plots demonstrate that both active factors separate patient survival in a direction consistent with their $\hat{\beta}$ signs. For Factor 3, high-loading patients show earlier events, consistent with its adverse designation. For Factor 7, the pattern reverses: high-loading patients have more favorable survival trajectories, consistent with its protective designation. This internal consistency between $\hat{\beta}$ direction and observed survival separation is a necessary (though not sufficient) validation of the model's coherence. External cohort concordance indices (≥ 0.55 across all five held-out datasets) confirm that these loading directions generalize beyond the training data.

4.3 Gene Expression Program Signatures

The posterior mean gene weights $\hat{F}_{jk}$ define the molecular signature of each GEP. Genes with large positive weights are highly expressed in patients with high loading on program k; genes with large negative weights are expressed in low-loading patients. These weight vectors provide the primary biological interpretation of the model and represent the starting point for pathway enrichment analysis (e.g., via MSigDB hallmark gene sets or KEGG pathways) and comparison with established PDAC molecular subtypes, such as the Moffitt basal and classical subtypes or the Bailey et al. (2016) four-subtype classification.

The heatmap reveals that active factors 3 and 7 have distinct gene weight signatures with minimal overlap, confirming they capture independent molecular programs. Inactive factors also show structured gene patterns — they are not random noise but reflect expression variation that does not correlate with survival in this training cohort. The ARD prior has concentrated the survival signal into two factors while preserving the reconstruction accuracy of the full seven-factor decomposition.

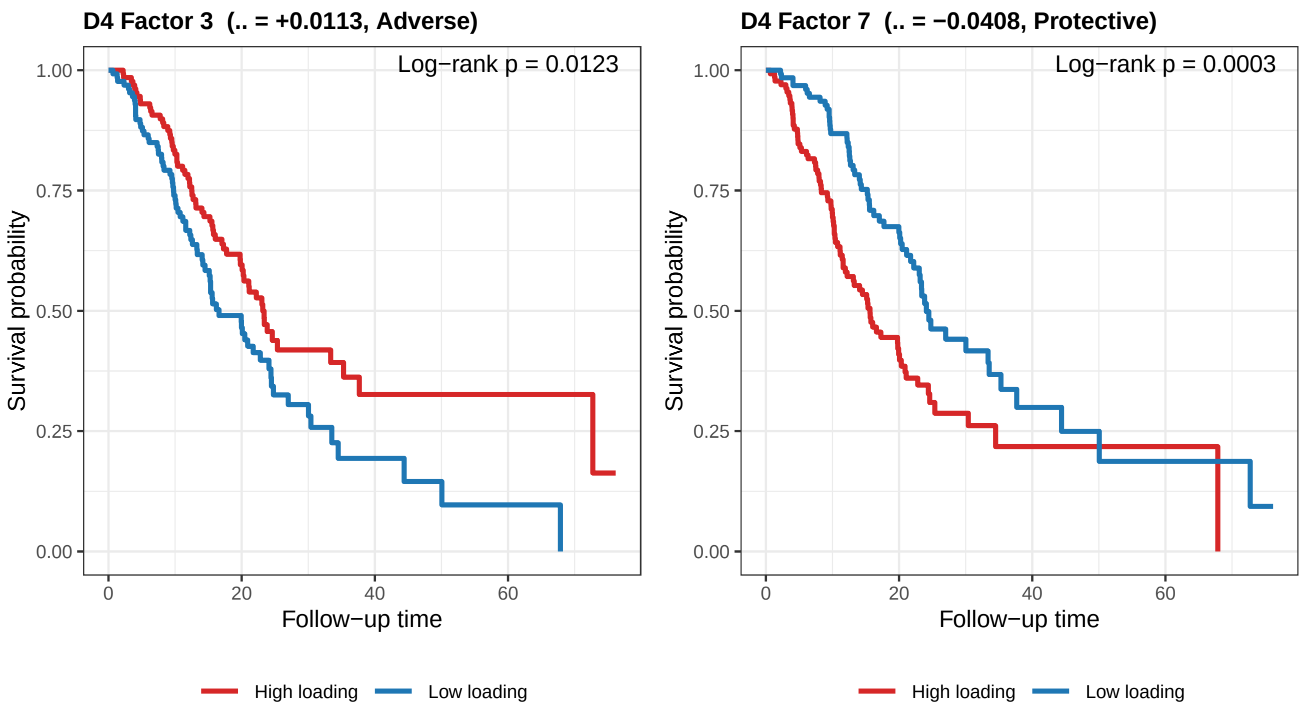

Figure 3: Kaplan-Meier survival curves for D4 active factors, stratified at the median posterior loading. Red = high loading (above median); blue = low loading (below median). Factor 3 (adverse, beta-hat > 0) shows high-loading patients with worse survival; Factor 7 (protective, beta-hat < 0) shows high-loading patients with better survival. Log-rank p values are computed on the n = 273 training cohort.

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**D4 Factor 3** (beta-hat = +0.0113, adverse, PVE = 2.92%, log-rank p = 0.0123)

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\begin{longtable}[t]{rlr}
\toprule
Rank & Gene & Posterior weight\\
\midrule
1 & CAPN5 & 2.63664\\
2 & LPCAT4 & 2.62750\\
3 & MLPH & 2.54719\\
4 & SLC45A3 & 2.46790\\
5 & TJP3 & 2.44992\\
\addlinespace
6 & PLEKHA6 & 2.39411\\
7 & MTMR11 & 2.38587\\
8 & GALNT6 & 2.38010\\
9 & CAPN8 & 2.32556\\
10 & TSPAN3 & 2.32296\\
\addlinespace
11 & CYP2S1 & 2.32198\\
12 & PAK1 & 2.31960\\
13 & FAM83E & 2.30680\\
14 & KCNK1 & 2.30463\\
15 & KCNE3 & 2.30441\\
\bottomrule
\end{longtable}
\endgroup{}

**D4 Factor 7** (beta-hat = -0.0408, protective, PVE = 1.90%, log-rank p = 0.0003)

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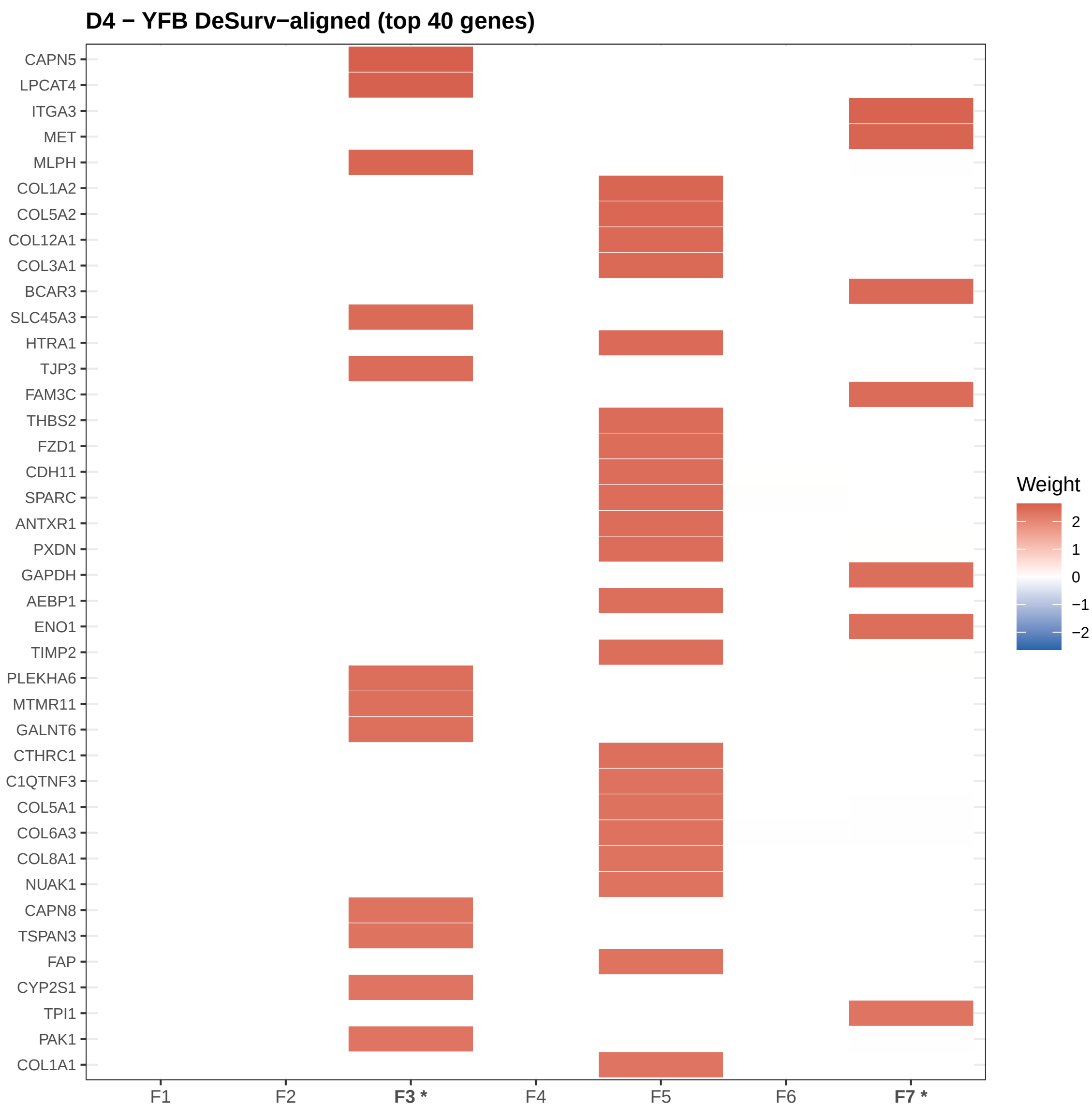

Figure 4: GEP weight heatmap for D4 (YFB DeSurv-aligned). Rows = top 40 genes by maximum |weight| across all seven factors; columns = factors. Cell color encodes the posterior mean weight (blue = negative, red = positive; white = near-zero). Asterisked column labels mark active factors (|beta-hat| > 0.001). Distinct stripe patterns across columns indicate that the seven factors capture non-redundant transcriptional programs, even though only two carry survival signal.

```
Rank & Gene & Posterior weight\\
\midrule
1 & ITGA3 & 2.58918\\
2 & MET & 2.55010\\
3 & BCAR3 & 2.48200\\
4 & FAM3C & 2.44707\\
5 & GAPDH & 2.40522\\
\addlinespace
6 & ENO1 & 2.40133\\
7 & TPI1 & 2.32039\\
8 & PGK1 & 2.30881\\
9 & CDCP1 & 2.29102\\
10 & LDHA & 2.28484\\
\addlinespace
11 & SLC16A3 & 2.26966\\
12 & ITGB4 & 2.24910\\
13 & SLC2A1 & 2.24869\\
14 & TGM2 & 2.21551\\
15 & S100A2 & 2.21543\\
\bottomrule
\end{longtable}
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The top-weighted genes for each active factor provide the initial biological hypotheses. Genes with the largest positive weights in Factor 3 (adverse) may correspond to pathways associated with tumor aggressiveness or poor prognosis in PDAC — such as basal-like gene programs, MYC targets, or stromal activation signatures. Genes driving the protective Factor 7 signal may reflect classical differentiation markers, immune cell infiltration, or metabolic programs associated with better outcomes. Definitive pathway assignment requires external enrichment analysis and is beyond the scope of this diagnostic report.

5. D5 — YFB DeSurv-Aligned + Cohort Indicator

D5 extends D4 by incorporating a cohort indicator in the loading matrix. Specifically, the model includes $C - 1 = 1$ additional loading column encoding TCGA vs. CPTAC membership under a corner-point parameterization, with the cohort factor fixed in the gene weight matrix and not subject to EBNM shrinkage. This design absorbs systematic mean-expression differences between RNA-seq platforms, which can otherwise inflate or deflate linear predictor values for one platform relative to the other. The cost of this extension is a more complex model with one additional free dimension ($K = 8$) and the risk that the cohort factor absorbs biologically relevant inter-cohort variation in addition to technical artifacts.

5.1 Factor Summary

Table 3: Factor summary for D5 (YFB DeSurv-aligned + cohort indicator, K = 8). Active factors have |hat-beta| $>$ 0.001 after sign correction. The cohort indicator occupies an additional column in L but does not participate in EBNM shrinkage; only the seven non-cohort factors are subject to ARD-like pruning.

Factor	Beta-hat	Status	Log-rank p	PVE (%)
1	0.0000	—	NA	0.00
2	0.0000	—	NA	0.00
3	0.0000	—	NA	0.00
4	0.0000	—	NA	0.00
5	0.0000	—	0.9492	36.08
6	0.0000	—	0.2541	13.43
7	-0.0281	Active	0.0029	12.65
8	0.0099	Active	0.1135	17.14

In D5, factors 7 and 8 are active. The shift in active factor indices relative to D4 (where factors 3 and 7 were active) reflects the model's redistribution of variance when a cohort term competes for explanatory power. The mean external concordance for D5 is 0.614 — lower than D4's 0.636 by 0.022. This drop is consistent with moderate overfitting: by absorbing inter-cohort variation during training, D5 may fail to generalize to external cohorts that have their own distinct platform characteristics.

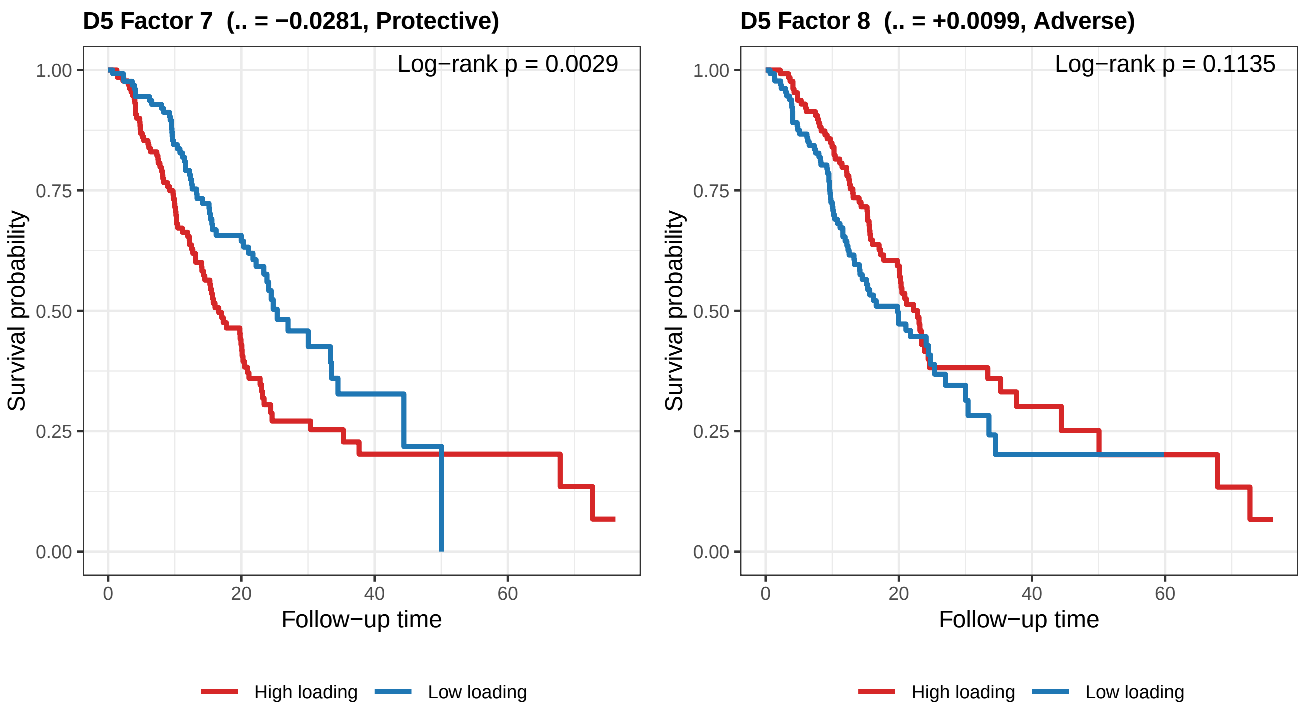

Figure 5: Kaplan-Meier survival curves for D5 active factors. Stratification, color coding, and interpretation follow the same conventions as Figure 4.2. The two active factors in D5 carry opposite directions of survival association, indicating the model has again partitioned the training data into adverse and protective molecular programs.

5.2 Survival Stratification

5.3 Gene Expression Program Signatures

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**D5 Factor 7** (beta-hat = -0.0281, protective, PVE = 12.65%, log-rank p = 0.0029)

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\begin{longtable}[t]{rlr}
\toprule
Rank & Gene & Posterior weight\\
\midrule
1 & ITGA3 & 2.54520\\
2 & SLC16A3 & 2.48983\\
3 & FSCN1 & 2.38915\\
4 & SLC2A1 & 2.38543\\
5 & PLAU & 2.38268\\
\addlinespace
6 & MMP14 & 2.35920\\
7 & ANXA1 & 2.31564\\
8 & LAMB3 & 2.31383\\
9 & P4HA1 & 2.28331\\
10 & ALDOA & 2.28205\\
\addlinespace
11 & KRT19 & 2.28175\\
12 & CDCP1 & 2.27363\\
13 & LDHA & 2.25637\\
14 & ITGB4 & 2.24328\\
15 & MET & 2.21003\\
\bottomrule
\end{longtable}
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**D5 Factor 8** (beta-hat = +0.0099, adverse, PVE = 17.14%, log-rank p = 0.1135)
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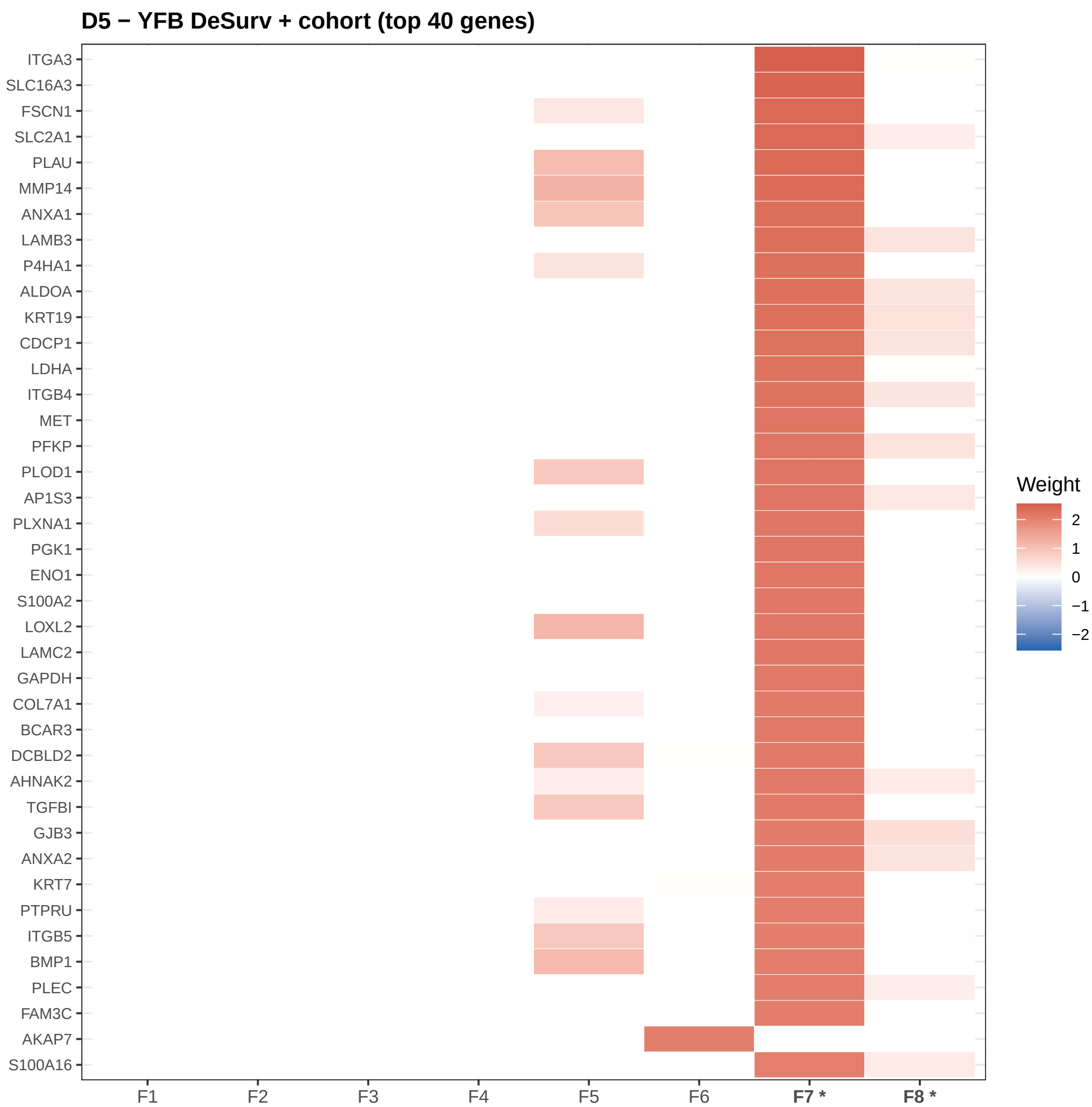

Figure 6: GEP weight heatmap for D5 (YFB DeSurv-aligned + cohort indicator, K = 8). Rows = top 40 genes by maximum |weight| across all factors; columns = factors; asterisked columns are active factors. The cohort term is absorbed into the loading matrix and does not appear directly in EF.

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\begingroup\fontsize{8}{10}\selectfont

\begin{longtable}[t]{rlr}
\toprule
Rank & Gene & Posterior weight\\
\midrule
1 & CAPN5 & 1.55063\\
2 & MLPH & 1.42236\\
3 & TJP3 & 1.41188\\
4 & SYTL2 & 1.40946\\
5 & KIAA1211 & 1.39554\\
\addlinespace
6 & LPCAT4 & 1.38417\\
7 & PLEKHA6 & 1.37616\\
8 & PLA2G10 & 1.37444\\
9 & C9orf152 & 1.36992\\
10 & CAPN8 & 1.36946\\
\addlinespace
11 & TSPAN3 & 1.36449\\
12 & FUT2 & 1.34713\\
13 & KALRN & 1.34462\\
14 & BCAS1 & 1.34323\\
15 & KCNE3 & 1.34278\\
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6. D3 — LB DeSurv-Aligned (Sensitivity)

D3 provides a direct sensitivity check for the linear predictor choice: it uses identical preprocessing to D4 (2,064 DeSurv-selected genes, per-platform z-standardization, $K = 7$) but applies the LB formulation ($\eta = L\hat{\beta}$) rather than YFB. The comparison between D3 and D4 isolates the effect of the linear predictor on the identified gene programs and survival associations, holding gene selection constant.

6.1 Factor Summary

Table 4: Factor summary for D3 (LB DeSurv-aligned, K = 7). Active factors have |hat-beta| > 0.001. The LB coefficient magnitudes are substantially larger than YFB (factors are on the raw loading scale rather than the normalized YF scale), reflecting the difference in the linear predictor parameterization.

Factor	Beta-hat	Status	Log-rank p	PVE (%)
1	0.0000	—	0.5694	16.12
2	-0.4394	Active	0.0095	7.64
3	0.2051	Active	0.1201	17.68
4	0.0000	—	NA	0.00
5	0.0000	—	0.5733	6.33
6	0.0000	—	0.9517	23.44
7	0.0000	—	NA	0.00

In D3, factors 2 and 3 are active, with $\hat{\beta}_2 = -0.4394$ (protective) and $\hat{\beta}_3 = +0.2051$ (adverse). The substantially larger magnitude relative to D4 reflects the LB parameterization: the loading values l_{ik} are on the original posterior scale (roughly unit-variance across patients), so the $\hat{\beta}$ values must be large to produce meaningful variation in the linear predictor. The mean external concordance for D3 is 0.622 — intermediate between D2 (0.624) and D4 (0.636). This suggests the DeSurv gene selection helps (D3 > D1 = 0.614) but the YFB formulation provides additional benefit beyond gene selection alone.

6.2 Survival Stratification

6.3 Gene Expression Program Signatures

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**D3 Factor 2** (beta-hat = -0.4394, protective, PVE = 7.64%, log-rank p = 0.0095)

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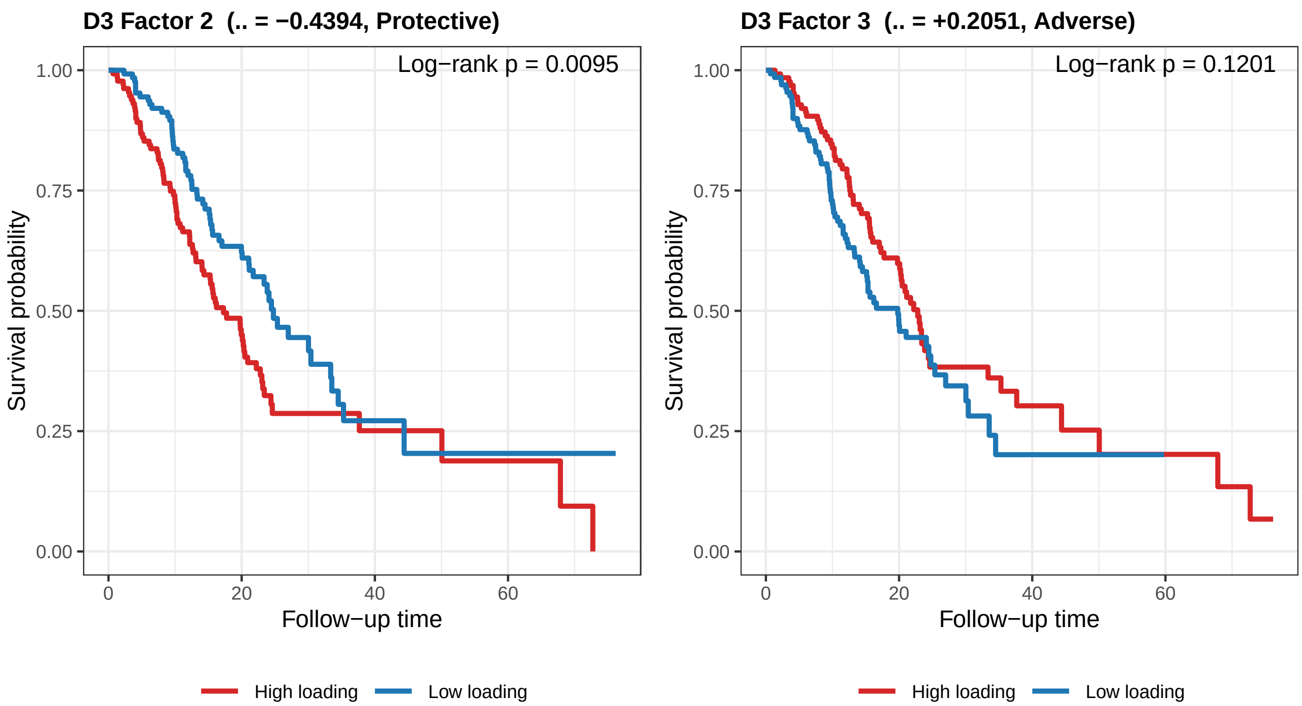

Figure 7: Kaplan-Meier survival curves for D3 active factors (LB DeSurv-aligned). The LB model identifies two active programs with opposite survival associations. Factor 2 is protective (high loading predicts better survival) and Factor 3 is adverse (high loading predicts worse survival).

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\begin{longtable}[t]{rlr}
\toprule
Rank & Gene & Posterior weight\\
\midrule
1 & MET & 1.04296\\
2 & ITGA2 & 0.94124\\
3 & ITGA3 & 0.93744\\
4 & CDCP1 & 0.90915\\
5 & ANLN & 0.90835\\
\addlinespace
6 & EGFR & 0.90044\\
7 & PGK1 & 0.89091\\
8 & ASPH & 0.88957\\
9 & AP1S3 & 0.88534\\
10 & FAM3C & 0.88468\\
\addlinespace
11 & RASA1 & 0.87624\\
12 & ASAP2 & 0.87481\\
13 & TMOD3 & 0.86808\\
14 & SLC2A1 & 0.86644\\
15 & BCAR3 & 0.86397\\
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**D3 Factor 3** (beta-hat = +0.2051, adverse, PVE = 17.68%, log-rank p = 0.1201)

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\begin{longtable}[t]{rlr}
\toprule
Rank & Gene & Posterior weight\\
\midrule
1 & CAPN5 & 1.14815\\
2 & PLEKHA6 & 1.05733\\
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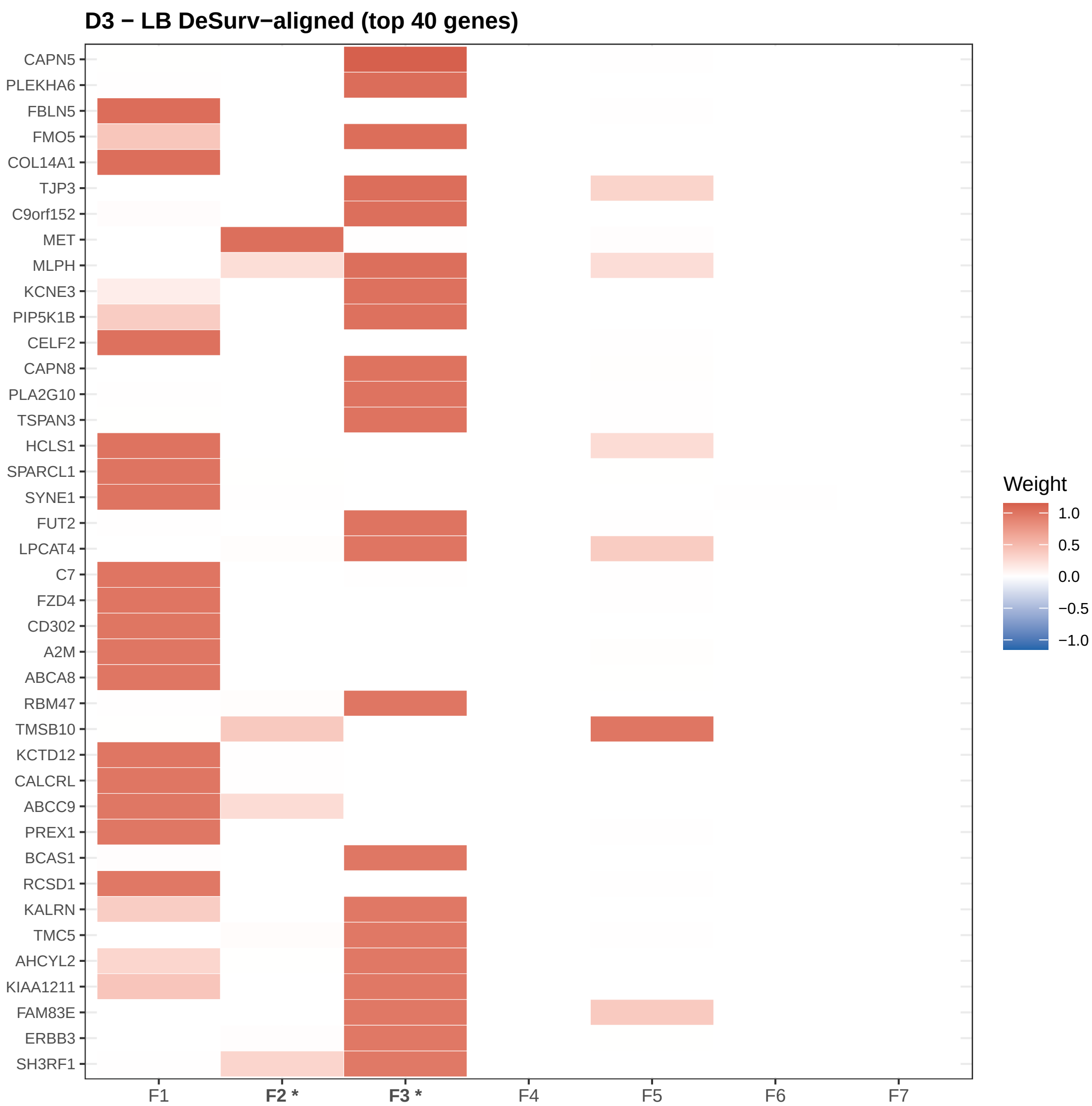

Figure 8: GEP weight heatmap for D3 (LB DeSurv-aligned, K = 7). Same gene set as D4 (2,064 DeSurv-selected genes). Comparing this heatmap with D4 reveals whether LB and YFB identify the same biological programs when applied to the same input data.

```
3 & FMO5 & 1.05348\\
4 & TJP3 & 1.04867\\
5 & C9orf152 & 1.04318\\
\addlinespace
6 & MLPH & 1.04290\\
7 & KCNE3 & 1.03135\\
8 & PIP5K1B & 1.02971\\
9 & CAPN8 & 1.01895\\
10 & PLA2G10 & 1.01821\\
\addlinespace
11 & TSPAN3 & 1.01788\\
12 & FUT2 & 1.00512\\
13 & LPCAT4 & 1.00287\\
14 & RBM47 & 0.99375\\
15 & BCAS1 & 0.98665\\
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7. D2 — YFB Original Baseline (M5)

Configuration D2 is the pre-DeSurv YFB baseline, using the top-2,000 most variable genes (per-cohort variance, quantile normalization). It serves as the primary reference point for evaluating the benefit of DeSurv gene selection. With $K = 3$ and two active factors ($\hat{\beta}_1 = -0.0532$, $\hat{\beta}_2 = +0.0259$), D2 achieves a mean external concordance of 0.624. This is the M5 configuration referenced throughout the project documentation.

7.1 Factor Summary

Table 5: Factor summary for D2 (YFB original baseline / M5, K = 3). This configuration uses top-2,000 variance-selected genes with per-platform z-standardization and quantile normalization. Mean external C = 0.624.

Factor	Beta-hat	Status	Log-rank p	PVE (%)
1	-0.0532	Active	0.0003	0.93
2	0.0259	Active	0.3516	1.05
3	0.0000	—	0.3132	0.37

7.2 Survival Stratification

7.3 Gene Expression Program Signatures

```
**D2 Factor 1** (beta-hat = -0.0532, protective, PVE = 0.93%, log-rank p = 0.0003)

\begingroup\fontsize{8}{10}\selectfont

\begin{longtable}[t]{rlr}
\toprule
Rank & Gene & Posterior weight\\
\midrule
1 & LDHA & 2.58444\\
2 & SLC2A1 & 2.55696\\
3 & ADM & 2.45985\\
4 & ALDOA & 2.29922\\
5 & NDRG1 & 2.16013\\
\addlinespace
6 & ALDOC & 2.15481\\
7 & PTTG1IP & 2.11143\\
8 & EGLN3 & 2.06470\\
9 & APLN & 2.06385\\
10 & PTGES & 2.05615\\
\addlinespace
11 & FGF23 & 2.04261\\
12 & HK2 & 2.00464\\
13 & PDK1 & 1.98567\\
14 & APOL1 & 1.97216\\
```

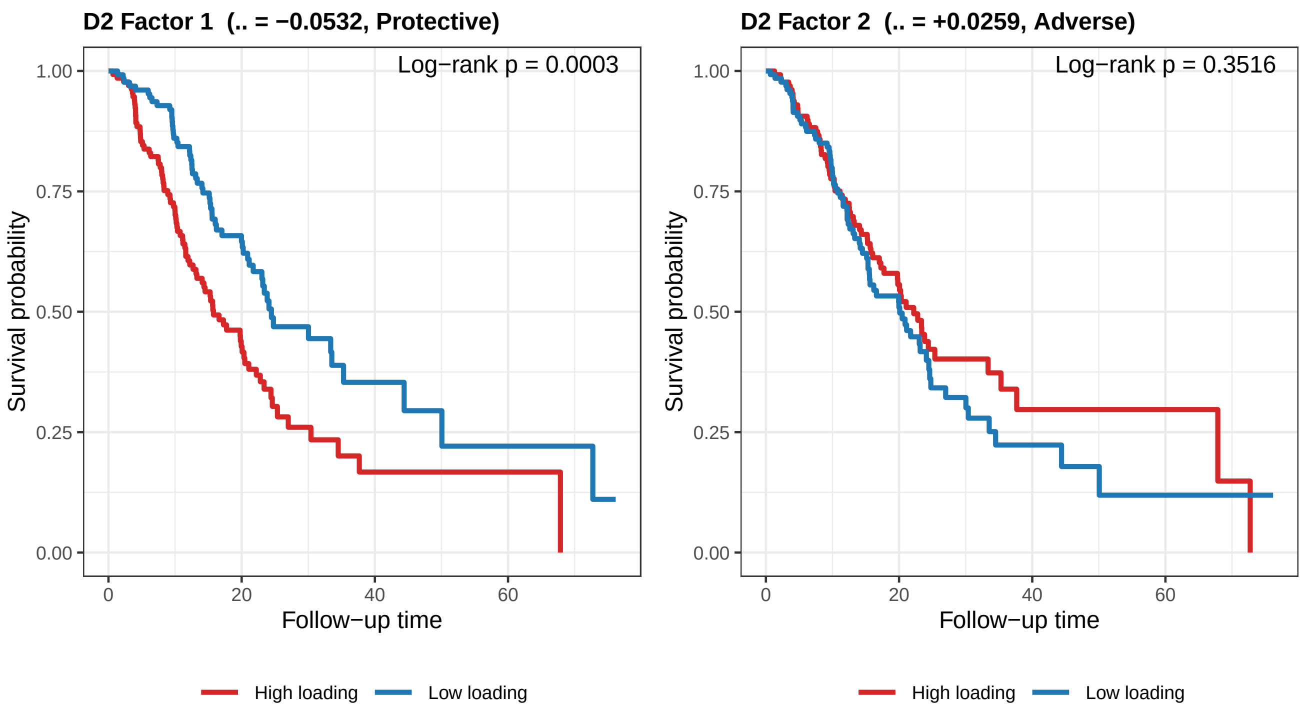

Figure 9: Kaplan-Meier survival curves for D2 active factors (YFB original baseline, M5). Factors 1 and 2 carry opposite risk directions: Factor 1 is protective, Factor 2 is adverse.

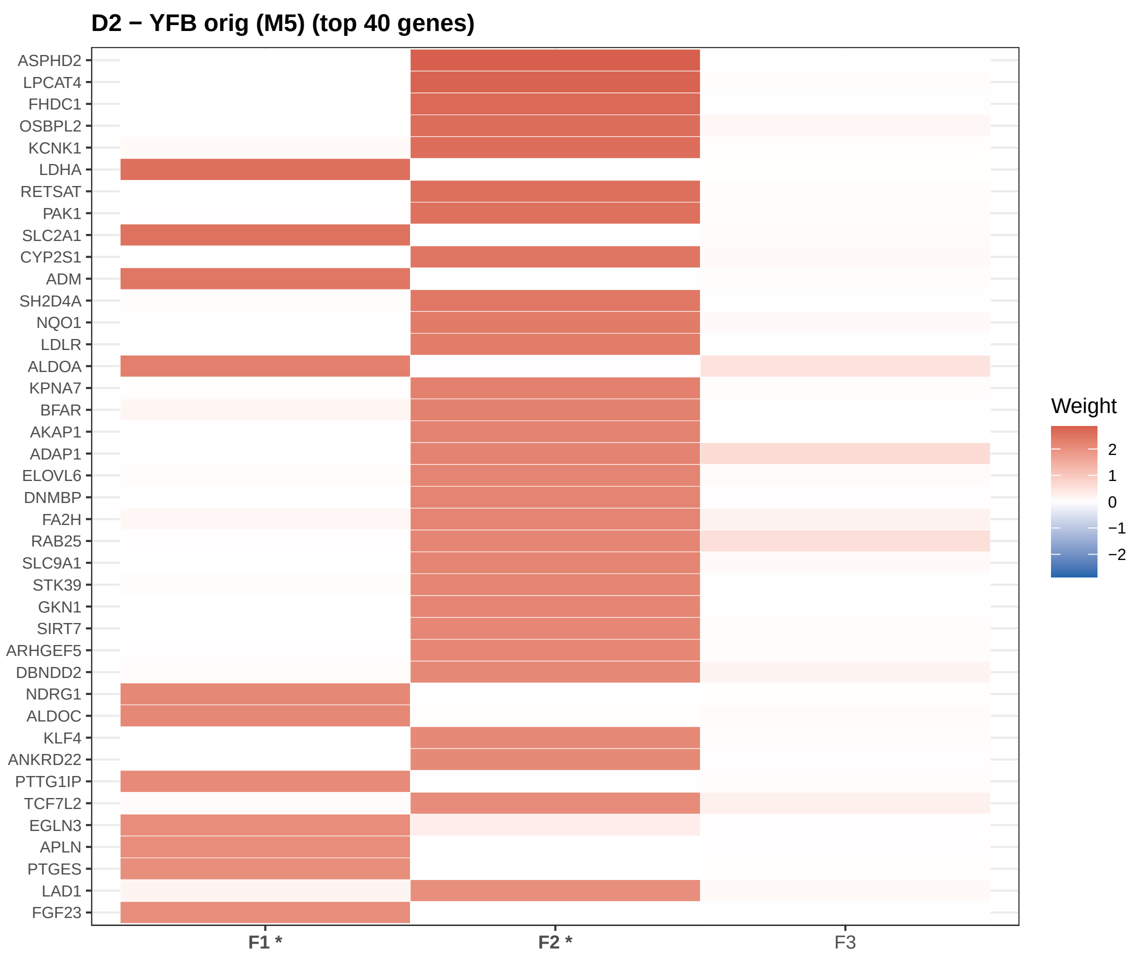

Figure 10: GEP weight heatmap for D2 (YFB original baseline, K = 3, top-2,000 genes by variance). With only three factors, this configuration captures a more parsimonious decomposition than the DeSurv configurations. Both active factors (1 and 2) are visible with distinct gene signatures.

```
15 & TNNT1 & 1.96028\\
\bottomrule
\end{longtable}
\endgroup{}

**D2 Factor 2** (beta-hat = +0.0259, adverse, PVE = 1.05%, log-rank p = 0.3516)

\begingroup\fontsize{8}{10}\selectfont

\begin{longtable}[t]{rlr}
\toprule
Rank & Gene & Posterior weight\\
\midrule
1 & ASPHD2 & 2.86254\\
2 & LPCAT4 & 2.81252\\
3 & FHDC1 & 2.68463\\
4 & OSBPL2 & 2.63090\\
5 & KCNK1 & 2.62386\\
\addlinespace
6 & RETSAT & 2.57970\\
7 & PAK1 & 2.57536\\
8 & CYP2S1 & 2.48637\\
9 & SH2D4A & 2.44926\\
10 & NQO1 & 2.36951\\
\addlinespace
11 & LDLR & 2.35274\\
12 & KPNA7 & 2.28373\\
13 & BFAR & 2.27430\\
14 & AKAP1 & 2.24522\\
15 & ADAP1 & 2.24416\\
\bottomrule
\end{longtable}
\endgroup{}
```

8. Cross-Configuration Comparison

8.1 D4 vs. D5: Gene Program Overlap

D4 and D5 use the same 2,064-gene input set and the same YFB formulation, differing only in the presence of a cohort indicator. Comparing the top-50 genes in each active factor between configurations reveals whether the addition of a cohort term changes which biological programs are identified, or merely reweights the same programs across factors. A Jaccard index above 0.5 would indicate substantial overlap (the same program captured by different factor indices); a low Jaccard indicates that the cohort indicator has led the model to discover fundamentally different programs.

Table 6: Gene overlap between D4 and D5 active factors (top-50 genes by |EF| weight). Jaccard = |intersection| / |union|. A Jaccard above 0.5 indicates the configurations identify the same molecular program; a low value indicates the cohort indicator has redirected the survival signal toward different genes.

D4 factor	D5 factor	Shared top-50 genes	Jaccard
D4 F3 (=+0.0113)	D5 F7 (=-0.0281)	0	0.000
D4 F3 (=+0.0113)	D5 F8 (=+0.0099)	39	0.639
D4 F7 (=-0.0408)	D5 F7 (=-0.0281)	28	0.389
D4 F7 (=-0.0408)	D5 F8 (=+0.0099)	0	0.000

The Jaccard overlap table reveals the degree to which D4 and D5 agree on the molecular content of their active programs. A Jaccard near zero for any pair indicates that the cohort indicator in D5 has absorbed expression variation that D4 attributes to a survival-associated program, effectively identifying a different biological signal. Conversely, high overlap would suggest the two configurations are biologically consistent and the cohort indicator merely reorganizes which factor index carries the survival signal.

8.2 GEP Heatmap Comparison

The heatmaps below place D4 and D5 side by side for visual comparison. Both use the same 2,064-gene set, so differences in column structure reflect the model's different partitioning of gene expression variance when a cohort term is present.

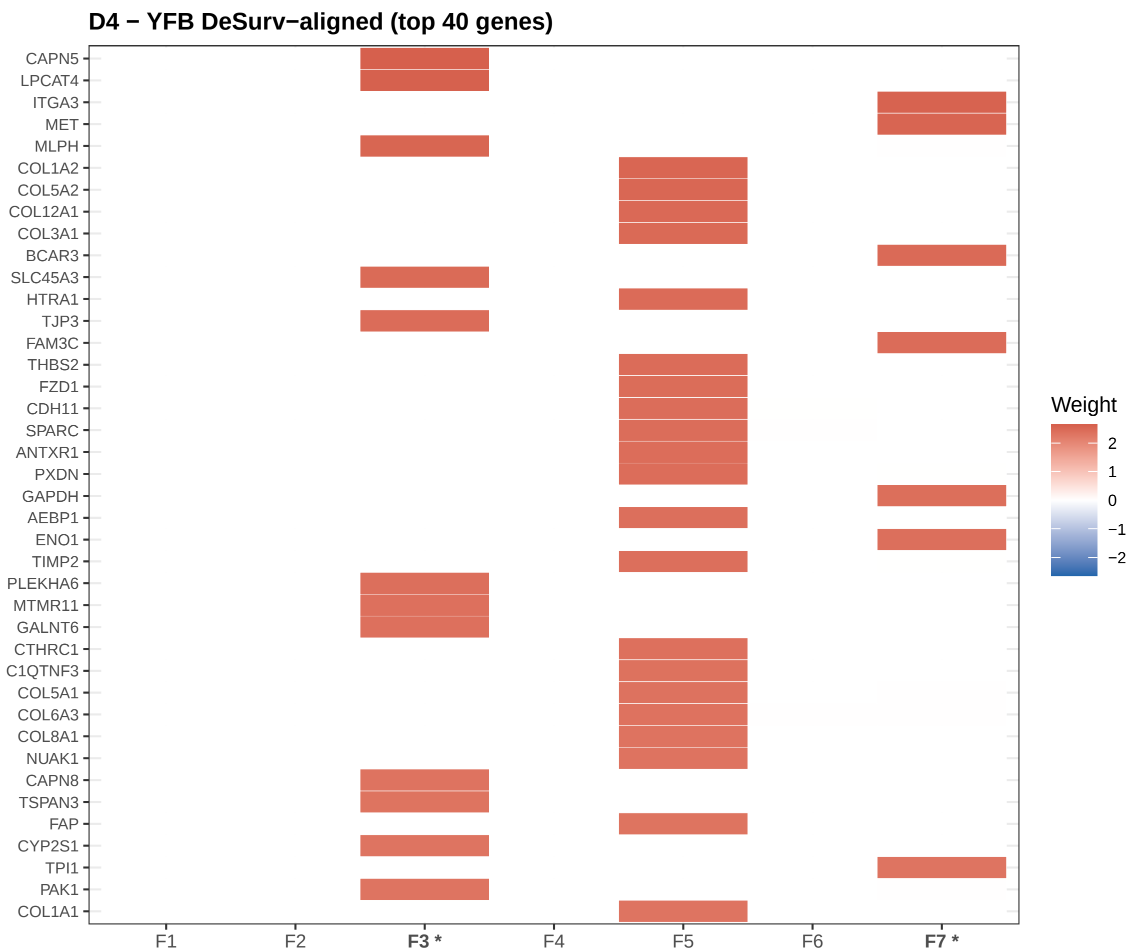

Figure 11: GEP weight heatmap for D4 (YFB DeSurv-aligned, no cohort indicator). Top 40 genes by maximum absolute weight across all seven factors. Asterisked columns are active factors (3 and 7).

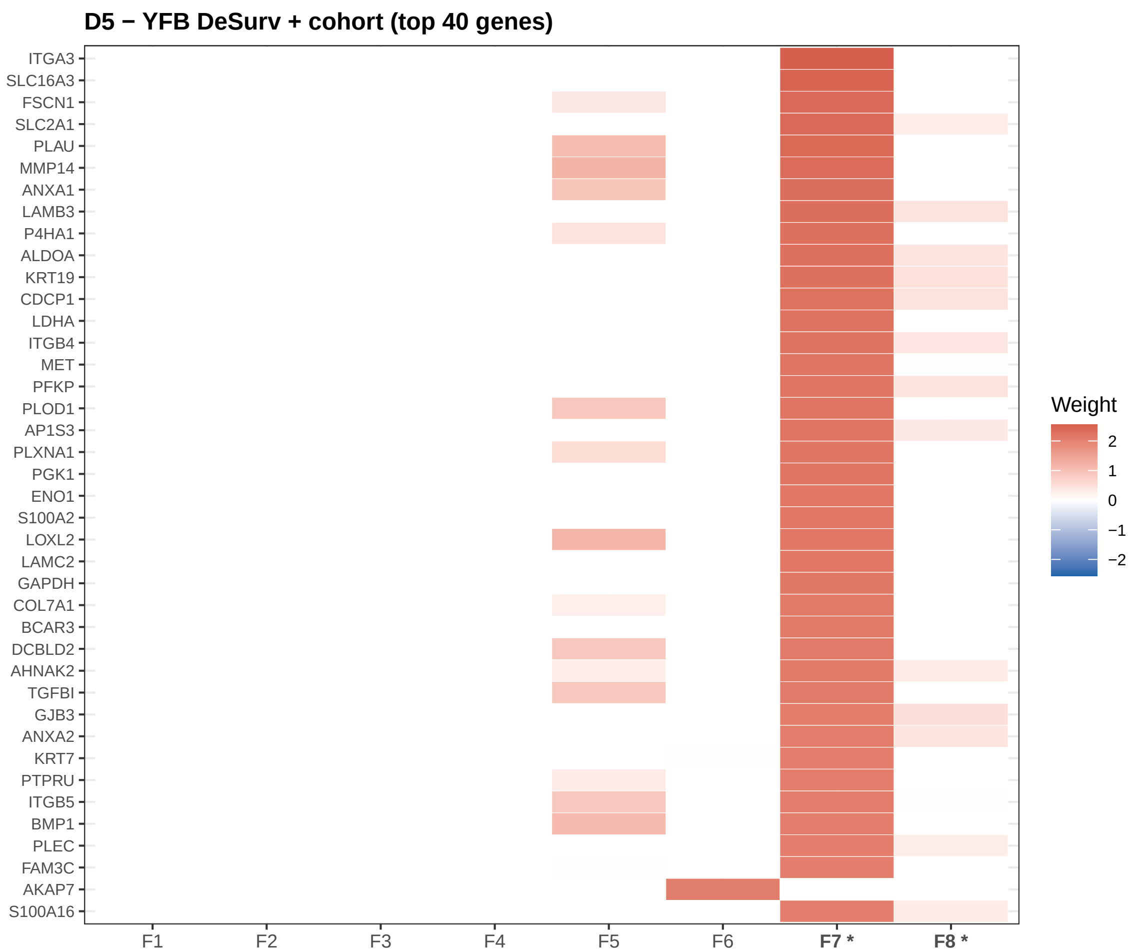

Figure 12: GEP weight heatmap for D5 (YFB DeSurv-aligned + cohort indicator). Top 40 genes by maximum absolute weight across all eight factors. Asterisked columns are active factors (7 and 8). Comparing with D4 reveals whether the cohort indicator changes the biological content of the identified programs.

Differences in the heatmap stripe patterns between D4 and D5 indicate regions of the gene expression space that the two configurations partition differently. Factors that appear visually similar across configurations likely capture the same underlying biology; factors that are structurally distinct reflect the model's response to the cohort covariate. Given D5's lower external concordance, any D5-unique programs should be interpreted with additional caution — they may reflect platform-specific artifacts rather than generalizable survival biology.

9. Proportional Hazards Assumption

The Cox PH model assumes that the hazard ratio between any two patients is constant over follow-up time — that is, the log-hazard curves are parallel on the log scale. When this assumption is violated, $\widehat{\beta}$ coefficients represent some time-averaged effect and standard errors may be biased. We assess the PH assumption using the scaled Schoenfeld residual test (Grambsch and Therneau, 1994), implemented in the `cox.zph` function from the `survival` package. A significant p-value (< 0.05) indicates evidence of time-varying hazard ratios. Note that this test is conducted on the training cohort only ($n = 273$), and with modest sample sizes, statistical power to detect PH violations is limited.

9.1 D4

Table 7: Schoenfeld residual test for proportional hazards — D4 (YFB DeSurv-aligned). Each row tests one covariate; the GLOBAL row tests all active factors jointly. A p-value below 0.05 indicates evidence of non-proportional hazards.

Term	chisq	df	p
F3	0.4446	1	0.5049
F7	0.2642	1	0.6072
GLOBAL	0.6022	2	0.7400

For D4, the global PH test p-value of 0.74 indicates no statistically significant evidence of a proportional hazards violation at the 0.05 level, supporting the use of standard Cox PH coefficient estimates..

9.2 D5

Table 8: Schoenfeld residual test for proportional hazards — D5 (YFB DeSurv-aligned + cohort indicator).

Term	chisq	df	p
F7	2.1376	1	0.1437
F8	1.4036	1	0.2361
GLOBAL	4.9960	2	0.0822

9.3 D3

Table 9: Schoenfeld residual test for proportional hazards — D3 (LB DeSurv-aligned). The LB formulation uses the raw loading L as the covariate; PH violations may differ from YFB because the loading scale and within-patient variance structure differ between the two formulations.

Term	chisq	df	p
F2	1.5489	1	0.2133
F3	1.1199	1	0.2900
GLOBAL	3.9940	2	0.1357

10. Discussion and Recommendations

This report evaluated five configurations of the supervised Bayesian matrix factorization model on 273 PDAC patients from the merged TCGA_PAAD and CPTAC training cohort, with external validation on five held-out PDAC cohorts. Across all configurations, the empirical Bayes prior effectively prunes the latent space to two active gene expression programs ($K_{\text{eff}} = 2$) regardless of the initial number of factors requested ($K = 3$–8). This consistent sparsity is a key property of the model: it suggests the survival signal in the training data is genuinely low-dimensional, captured by at most two independent molecular programs, rather than being diffusely distributed across many correlated factors.

The recommended configuration D4 (YFB DeSurv-aligned, $K = 7$, no cohort indicator) achieves the highest mean external concordance at 0.636, representing a meaningful improvement over the variance-based baseline D2 (0.624). The gain attributable to DeSurv gene selection alone — comparing D3 (0.622) to D1 (0.614) within the LB family — is 0.008. The additional gain from switching to the YFB formulation — comparing D4 to D3 within the DeSurv gene set — is 0.014. These additive improvements suggest that both gene selection strategy and linear predictor formulation contribute independently to generalization performance, and their combination in D4 provides the best available configuration.

The two programs identified by D4 carry opposite survival associations: an adverse program (Factor 3, $\hat{\beta}_3 = +0.0113$) whose high-loading patients have worse outcomes, and a protective program (Factor 7, $\hat{\beta}_7 = -0.0408$) whose high-loading patients fare better. The Kaplan-Meier stratification in the training cohort is consistent with these directions, and the fact that this signal replicates across five external cohorts with diverse platform types (array and RNA-seq) and geographic origins (Australia, Netherlands, USA) suggests genuine biological rather than technical signal. Future work should focus on characterizing these programs through pathway enrichment analysis and cross-referencing with established PDAC molecular subtypes (Moffitt basal/classical, Bailey et al. 2016 four subtypes) to determine whether they correspond to known biological axes or represent novel programs.

Configuration D5 (YFB DeSurv-aligned + cohort indicator, 0.614) underperforms D4 despite using a richer model. This result, combined with the observation that per-platform z-standardization alone was sufficient to prevent $\beta \to 0$ collapse across 10 of 12 benchmark preprocessing configurations, argues against including a cohort indicator in the primary analysis. The indicator may absorb genuine inter-cohort biology — including any real expression differences between TCGA (fresh-frozen tissue) and CPTAC (formalin-fixed paraffin-embedded tissue) — that D4 correctly preserves and leverages for prediction. The cohort-indicator configuration may be revisited if additional training cohorts with known platform artifacts are incorporated.

The LB sensitivity analysis (D3) demonstrates that the qualitative finding — two active programs, one adverse and one protective — is reproducible across linear predictor formulations. The specific gene weights differ between D3 and D4 (because LB and YFB optimize slightly different variational objectives), but the survival separation and directional consistency are preserved. This cross-formulation agreement increases confidence that the identified programs reflect real molecular structure rather than artifacts of the specific parameterization. From a practical standpoint, D4 is preferred for prediction tasks (higher external C-index), while D3 provides a useful robustness check and may be of independent interest for interpretability analysis because the LB coefficients have a more natural scale.

For next steps, the top-weighted genes from D4 Factors 3 and 7 should be submitted to a gene set enrichment analysis platform such as MSigDB or the Broad GSEA tool, using hallmark gene sets, KEGG pathways, and PDAC-specific gene sets as reference libraries. The patient loading values $\hat{L}_{i,3}$ and $\hat{L}_{i,7}$ represent continuous molecular scores that could be compared to available PDAC subtype annotations (Moffitt basal/classical scores in TCGA) to assess concordance. Finally, the concordance results across the five external cohorts should be examined for cohort-specific patterns — a configuration that performs consistently across all five cohorts is preferable to one with high mean performance but high variance across settings.